Number Theory

Reflection

This was more like an extension to the number theory related units that I learned in SIT192 – Discrete Maths.

Basics

Congruence

When a, r, m ∈ $\mathbb{Z}$, and m > 0, and if m divides (a - r) we can say the statement below. This basically means that a and r leave the same remainder when divided by m.

a = r mod m *when* **m | (a - r)**

To reduce a number modulo m, just divide it by m and keep the remainder. For example,

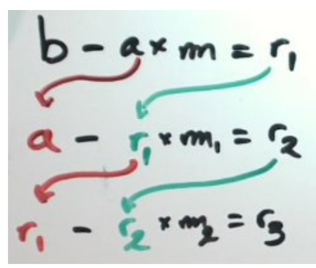
-27 ≡ **34 (mod 61)**, because 34 - (-27) = 61.

Euclidean algorithm

gcd(a,b) is the largest positive integer that divides both a and b. Two numbers are said to be relatively prime when their gcd is 1. This check is important for modular inverse, Fermat's Little Theorem, Euler's Theorem and the Chinese Remainder Theorem down the line.

When b > a, do this until $r_{i+1} = 0$. Then, r_i is the gcd.

$$\begin{aligned} \operatorname{rem}(b \div a) &= r_1 \\ \operatorname{rem}(a \div r_1) &= r_2 \\ \operatorname{rem}(r_1 \div r_2) &= r_3 \end{aligned}$$



Bezouts Lemma

If gcd(a, b) = c, then, **m ⋅ a + n ⋅ b = c**

a and b are relatively prime only if numbers m and n suffice **m ⋅ a + n ⋅ b = 1**

Instead of trying to guess m and n by substituting a and b, just re-arrange the equations. Use the b – a*m = r_1 method (Eucledian algorithm), and substitute values from where necessary and keep on doing that until the final equation comes to the form said above, with m and n being real integers.

Modular Inverse

The modular inverse of a modulo m is a number x such that: **a * x ≡ 1 (mod m)**. A modular inverse exists if and only if **gcd(a, m) = 1** (aka. they are relatively prime).

p ≡ m has an inverse (mod p)

q ≡ gcd (m,p) = 1

$p \longleftrightarrow q$

By Bezouts' ⇒

if the gcd (m,p) = 1,

there are 2x numbers 'a' and 'b':

$am + bp = 1$

we use extended euclidean algorithm

$\underbrace{am-1}$ = -bp

↳ a is a multiple of p

↳ $\underbrace{am}$ ≡ 1 mod p

inverse

To find the inverse of a **89 mod 28**, find m and n (Bezouts coefficients) such that **89m + 28n = 1**. So in this case, it would be **1 = -11*89 + 35*28**. This means that -11 is an inverse of **89 mod 28**. Also note that finding just the Bezouts coefficient of **89** (-11 in this case) is enough just for the sake of solving these quickly. As -11 is out of the 0-27 range (for mod 28), we will reduce it to get **-11+28 = 17**. Now, you can say that the inverse of 89 is **17 (mod 28).**

Solving Linear Equations in Modular Arithmetic

If gcd(m, p) = 1, then for any number x the equation mx = a (mod p) has a solution.

To solve a *x ≡ b (mod m), first check gcd(a, b). If this is equal to 1, find the inverse of **a** and multiply both sides by it. For example, let's look at **70x** ≡ **41 (mod 61)**:

- Reduce 70 mod 61 —> 70 ≡ 9 (mod 61)
- Use the extended euclidean algorithm —> 1 = 4*61 - 27*9
- The inverse of 9 is -27 ≡ 24 (mod 61)
- Multiply both sides by 34: x ≡ 34 * 41 = 1394 ≡ 52 (mod 61).

Or when generalized, it will look as follows, to solve ax ≡ b (mod p) if gcd(a, p) = 1,

- Find the inverse s of a modulo p using the extended Euclidean algorithm.
- Multiply both sides of the equation by s: asx ≡ bs (mod p).
- This results in the solution x ≡ bs (mod p).

The Chinese remainder theorem

This theorem lets us combine two modular equations into one equation, as long as the two moduli have no common factor other than 1. In other words, the gcd of the two moduli should be 1 to apply this algorithm.

For the two equations:

$x \equiv a \pmod{m}$

$x \equiv b \pmod{n}$

First, check if gcd(m,n)=1. If it is, proceed. Then, use the extended Euclidean algorithm to find m(s) and n(t).

$m*s + n*t = 1$

Using these values, substitute them to the final formula to get the solution.

$x \equiv bms + ant \pmod{mn}$

When there are three equations, solve the first two and then solve it again using it's solution and the third equation.

Fermat's Little Theorem

When p is prime and p does NOT divide a, Fermat's Little Theorem says that:

$$\mathbf{a^{p-1} \equiv 1 \pmod{p}}$$

This lets us use a large exponent modulo p-1. For example, let's look at $\mathbf{11^{27699} \bmod 13}$.

- $11^{12} \equiv 1 \pmod{13}$, because 13 is prime
- And, 27699 = 12 * 2308 + 3
- So,
  - $11^{27699} \equiv (11^{12})^{2308+3}$
  - $11^{27699} \equiv (11^{12})^{2308} \times 11^{3}$
  - $11^{27699} \equiv 11^{3}$
  - $11^{27699} \equiv 5 \pmod{13}$

However, if the p is not a prime number, you will have to use another method using the Euler's Totient Function. Only the first part changes, the rest of it remains almost identically.

Proof of $a^p \equiv a \pmod p$ —>

$a^p \equiv a \pmod p$ — p is prime, a is X a multiple of p

1. $k \cdot a = r_k \pmod p$ — for every k from 1 to p-1

2. we don't get $r_k = 0$,
   because $(k \cdot a - r_k)$ is a multiple of p
   $k \cdot a$ is not a multiple of p (as p is prime)
   $k \cdot a$ " " " " " (as its smaller than p)

why $r_k \neq r_j$ if $k \neq j$?

$$[\underbrace{(k-j)}_{} \cdot a - \underbrace{(r_k - r_j)}_{}]$$

not a multiple of p

r_k's take all values between 1 & p-1

3. multiply on both sides

$$\prod_{k=1}^{p-1} k \cdot a \equiv \prod_{k=1}^{p-1} r_k \pmod p$$

$$a^{p-1} \prod_{k=1}^{p-1} k \equiv \prod_{k=1}^{p-1} \pmod p$$

4 − $a^{p-1} \equiv 1 \pmod p$

$a^p \equiv a \pmod p$

Euler's Totient Function: ϕ

ϕ(n) counts the integers from 1 to n-1 that are relatively prime to n. To calculate this, either list out all the numbers from 1 to n-1 (excluding n and 0) and count the relatively prime numbers to n manually.

Or, you can break n down into it's prime factors (powers), and use this algorithm.

If $n = p_1^{a1} \times p_2^{a2} \times \ldots \times p_k^{ak}$

Then, $\phi(n) = n * (1 - \frac{1}{p1}) * (1 - \frac{1}{p2}) \ \ldots \ (1 - \frac{1}{p2})$

Perform prime factorization for n, take each prime factor p (without the powers) and multiply n by $(1 - \frac{1}{p})$.

Euler's Theorem

When gcd(a, n) = 1, Euler's theorem says:

$$\mathbf{a^{\phi(n)} \equiv 1 \pmod n}$$

Fermat's Little Theorem is the prime version of Euler's Theorem. This is because ϕ(p) = p-1 when p is prime. To simplify a large power, divide the exponent by ϕ(n) and keep the remainder

Examples

Fermat's Little Theorem & Euler's Totient Function

3. 7^{31317} mod 10 — not prime

$\gcd(7,10) \Rightarrow 10 - 7\times 1 = 3 \Rightarrow 1$
$7 - 3\times 2 = 1$

$\phi \Rightarrow \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\therefore \phi(10) = 4$

Eulers theorem $\Rightarrow 7^4 \equiv 1 \mod 10$

$31317 = 4 \times 7829 + 1$

$$7^{31317} = 7^{4(7829)+1}$$
$$= (7^4)^{7829} \times 7$$
$$\equiv 1^{7829} \cdot 7 \mod 10$$
$$\equiv 7 \mod 10$$

Exercise 1.4.4. Calculate:

1. 11^{27699} mod 13 — prime

$a^p \equiv a \pmod p$

$\gcd(11,13) \Rightarrow 13 - 11\times 1 = 2 \Rightarrow 1$
$11 - 2\times 5 = 1$

$11^{13-1} \equiv 1 \mod 13$
$11^{12} \equiv 1 \mod 13$

$27699 \div 12 \times 2308 = 3$
$27699 = 12\times 2308 + 3$

$$\therefore 11^{27699} = 11^{12\cdot 2308 + 3}$$
$$= (11^{12})^{2308} \times 11^3$$
$$\equiv 1^{2308} \cdot 11^3 \mod 13$$
$$\equiv 11^3 \mod 13$$
$$\equiv 5 \mod 13$$

$11^3 \equiv (-2)^3$
$\equiv -8$
$\equiv 5 \mod 13$

Finding the Last Digits of a Large Power

For the last k digits of a number n, calculate: **n mod 10^k**.

- For the last digit: **mod 10**.
- For the last two digit: **mod 100**.
- For the last three digit: **mod 1000**.